

Batch effects and confounders

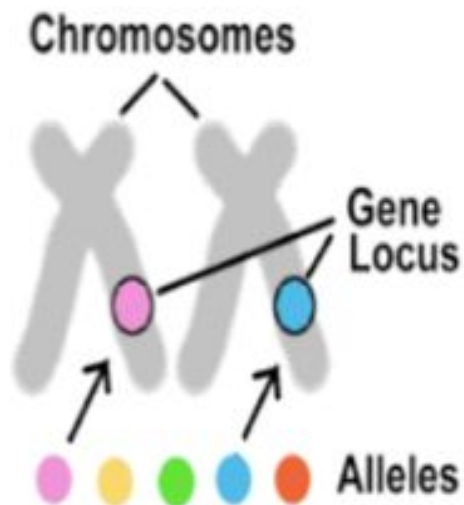
Jeff Leek

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**External Factors
(like environment)**

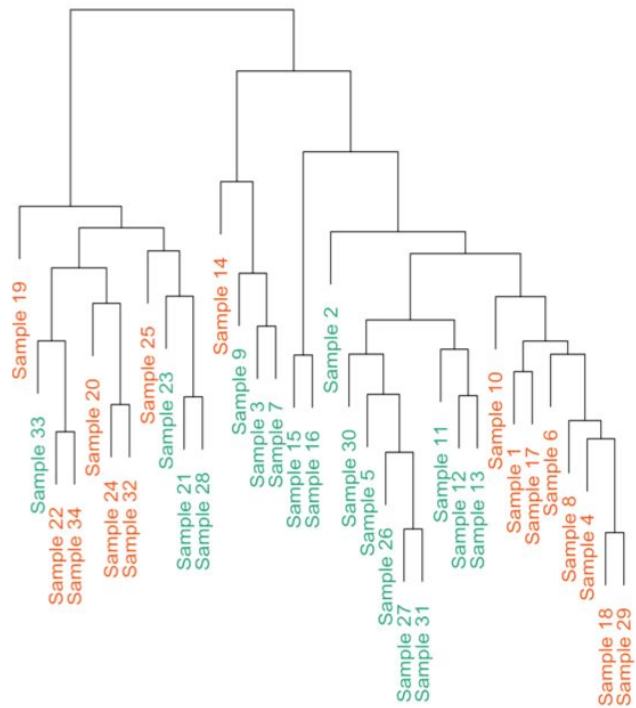


Genetics/Epigenetics

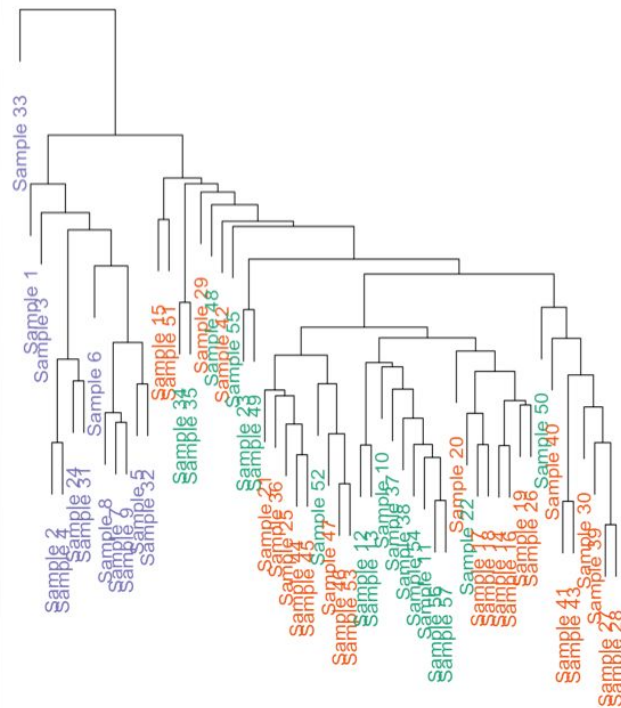


Technical Factors

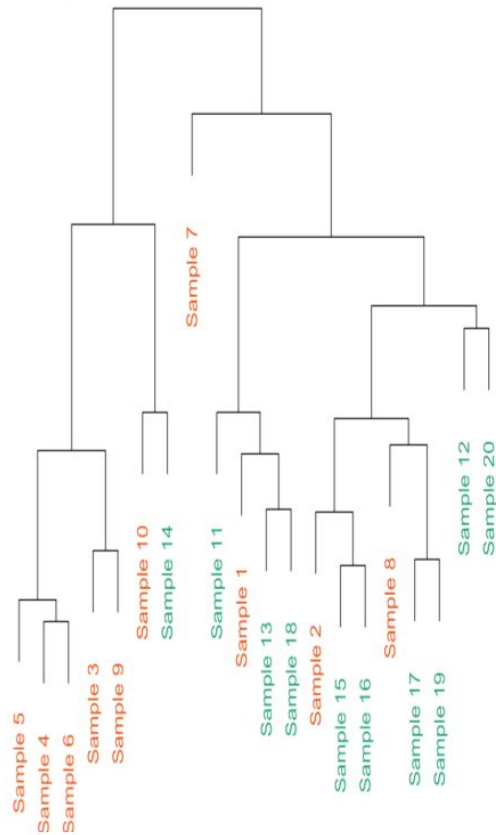
Color = Environment
(Idaghdour et al. 2008)

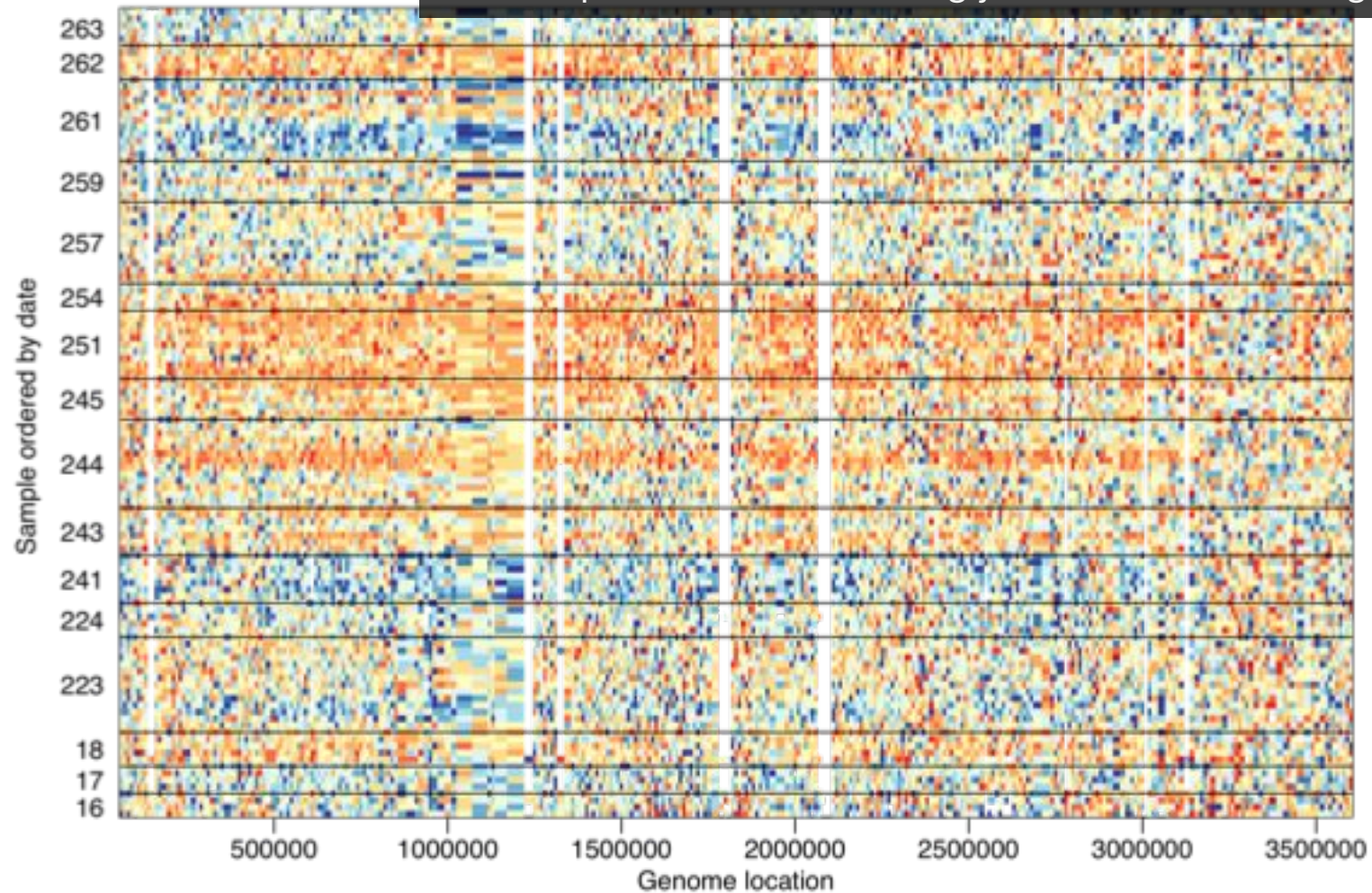


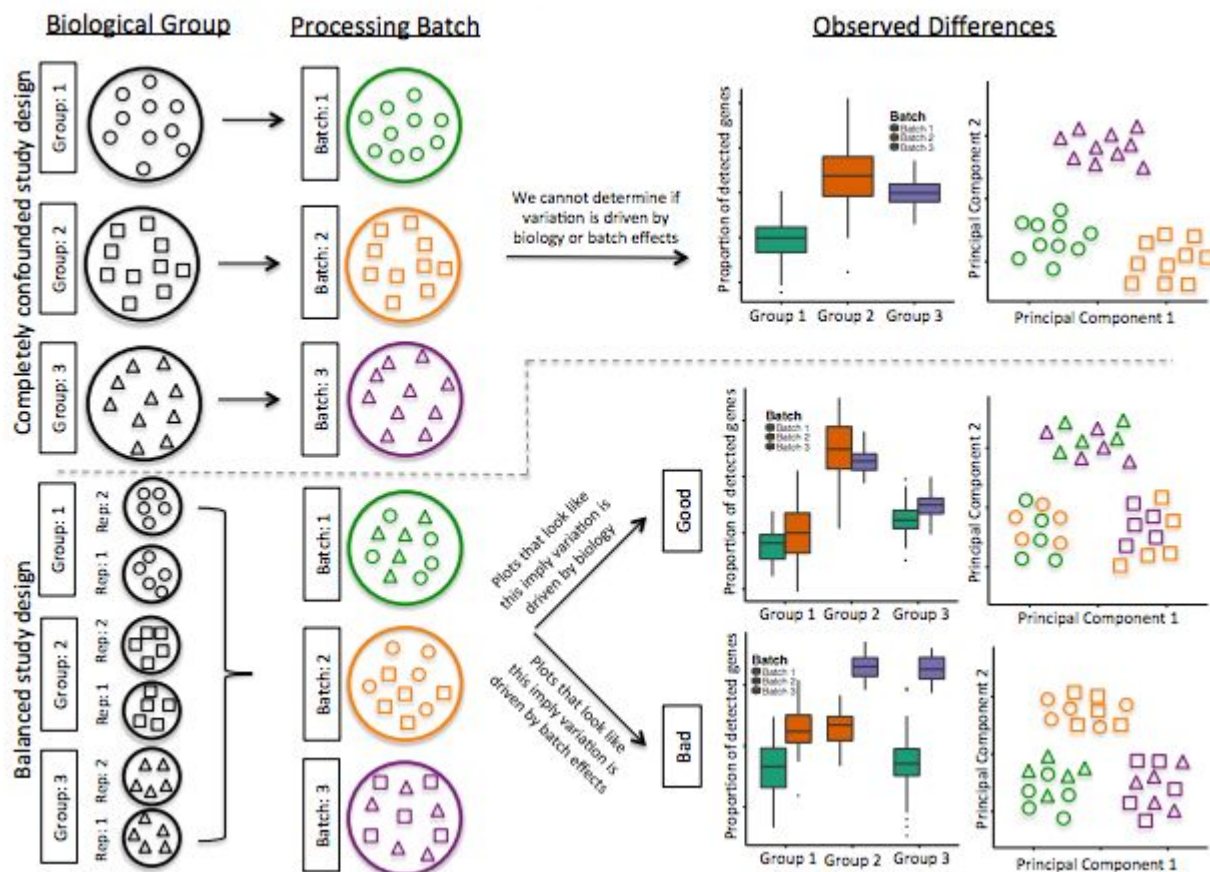
Color = Processing Year
(Cheung et al. 2008)



Color = Allele
(Brem et al. 2005)



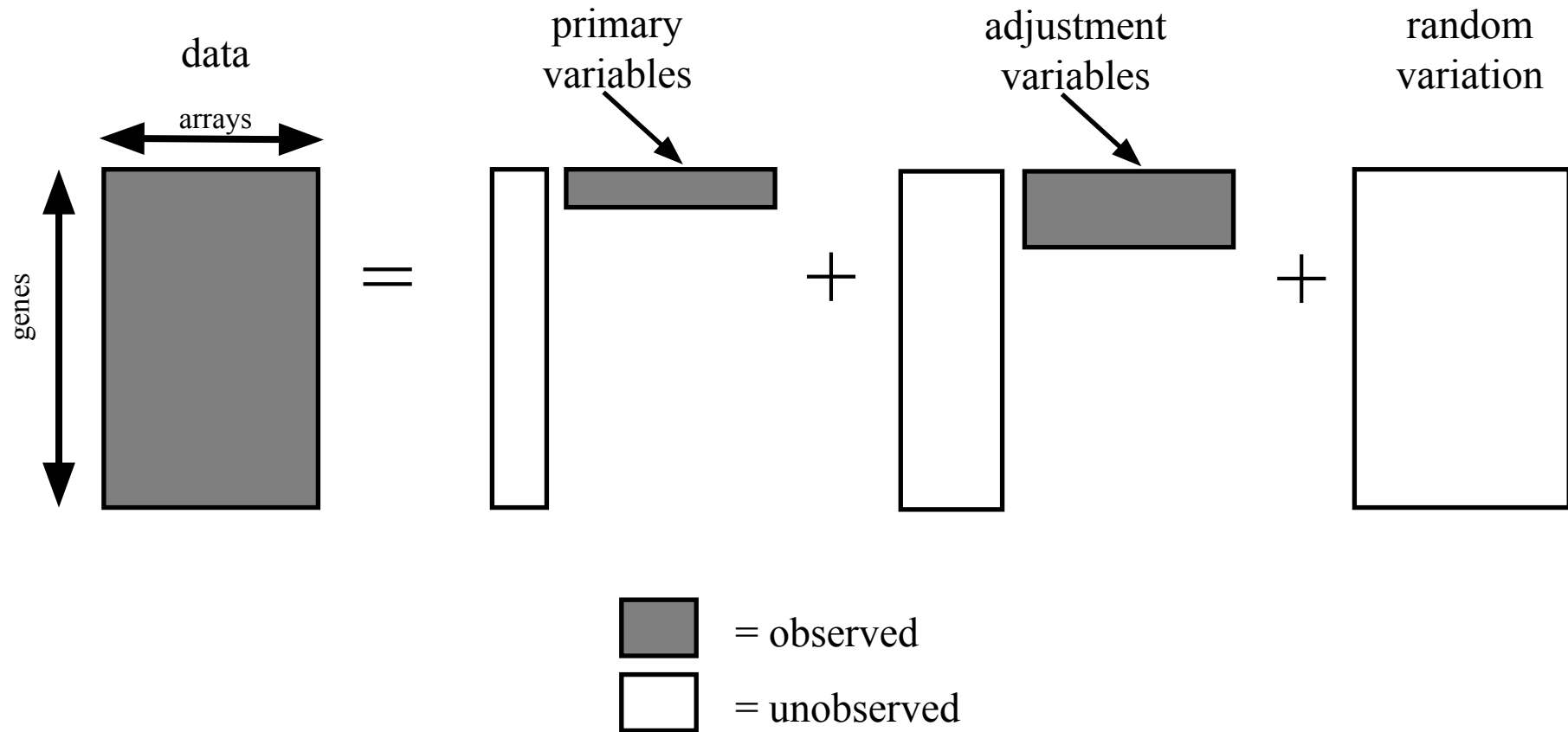




$$Y = b_0 + b_1 P + b_2 B + e$$

P = Phenotype you care about

B = Batch



Adjusting batch effects in microarray expression data using empirical Bayes methods

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Abstract

Non-biological experimental variation or "batch effects" are commonly observed across multiple batches of microarray experiments, often rendering the task of combining data from these batches difficult. The ability to combine microarray data sets is advantageous to researchers to increase statistical power to detect biological phenomena from studies where logistical considerations restrict sample size or in studies that require the sequential hybridization of arrays. In general, it is inappropriate to combine data sets without adjusting for batch effects. Methods have been proposed to filter batch effects from data, but these are often complicated and require large batch sizes (> 25) to implement. Because the majority of microarray studies are conducted using much smaller sample sizes, existing methods are not sufficient. We propose parametric and non-parametric empirical Bayes frameworks for adjusting

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Biostat (2007) 8 (1): 118-127.
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First published online: April 21, 2006

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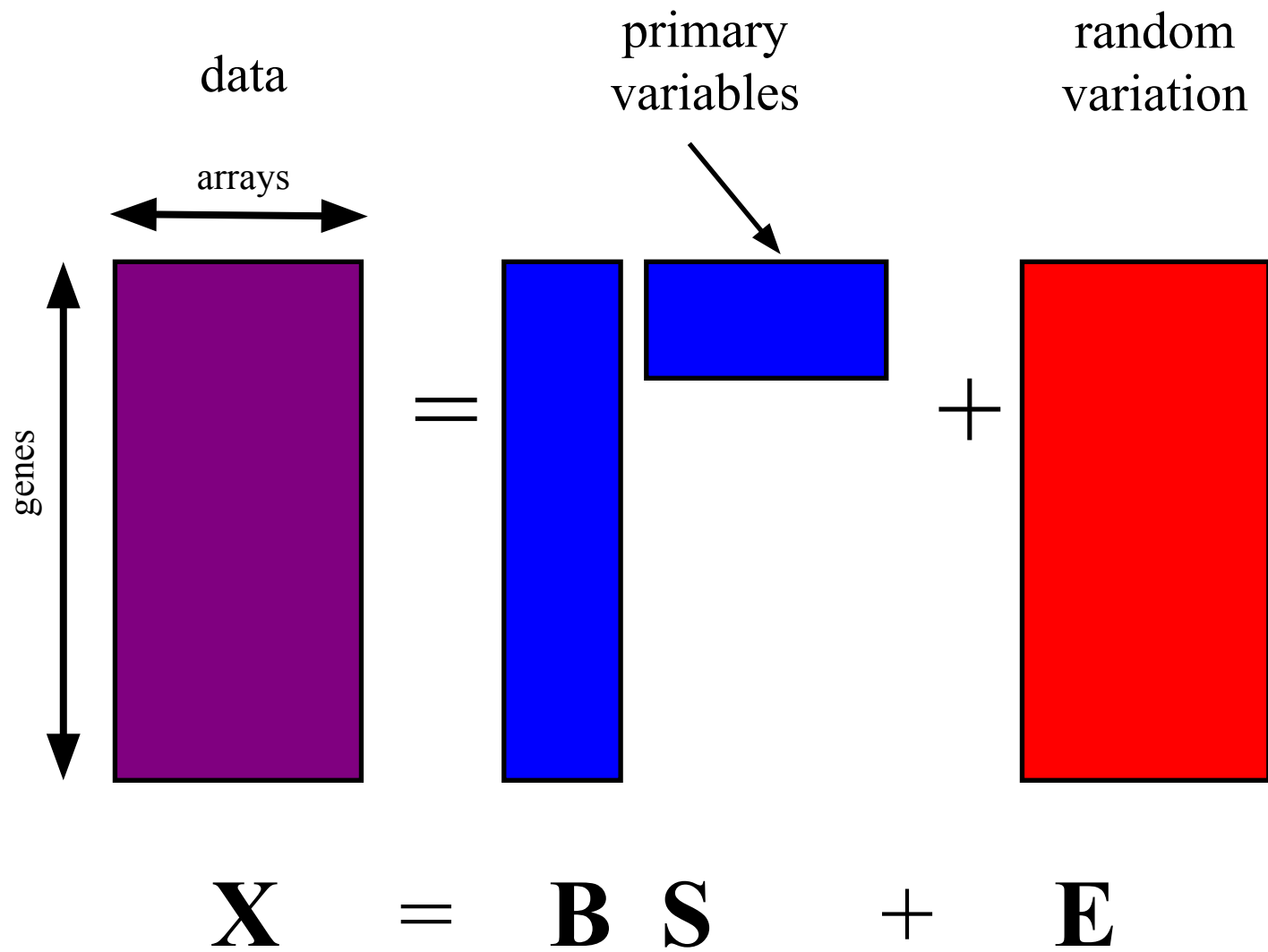
Current Issue

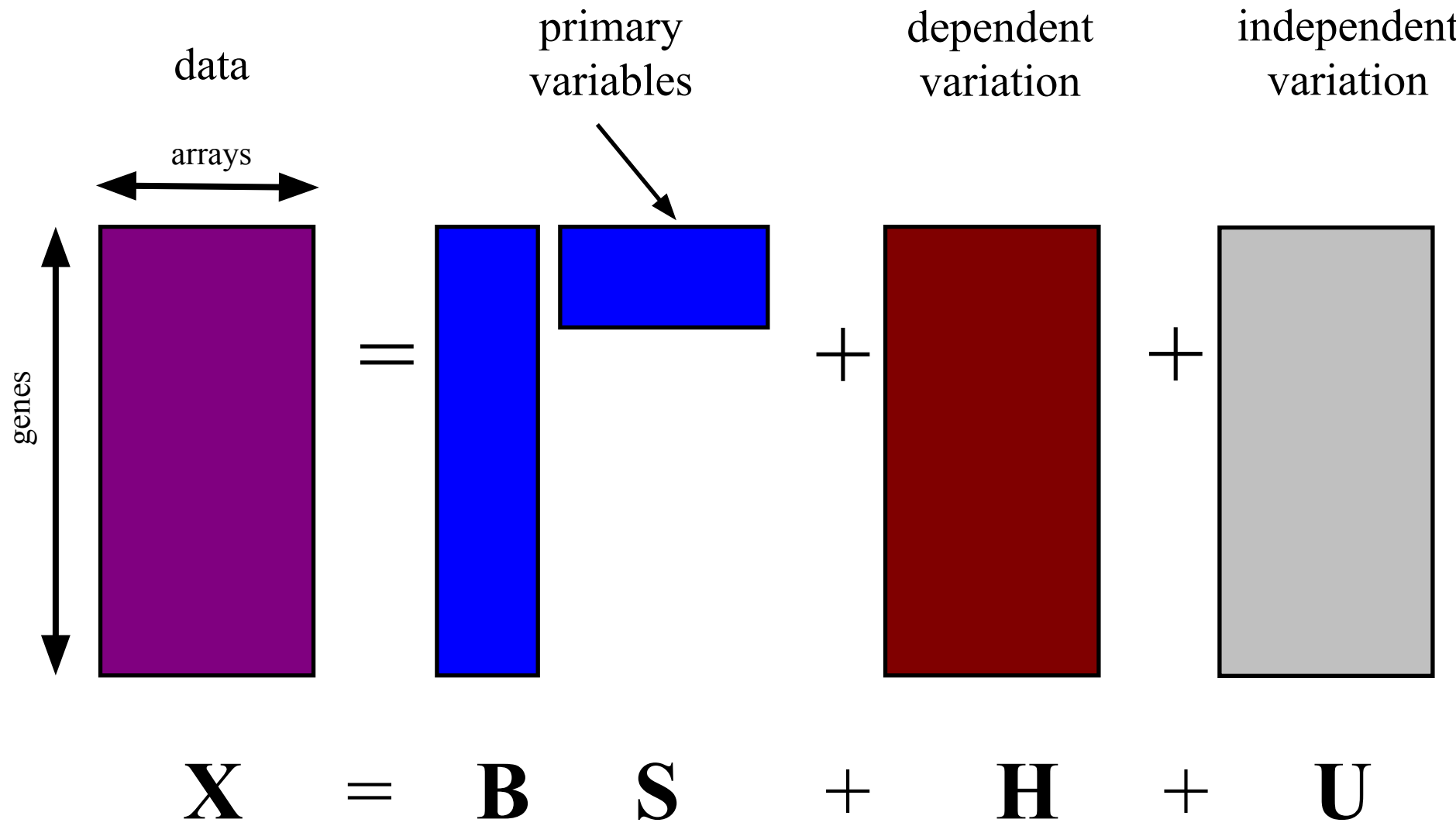
July 2015 16 (3)

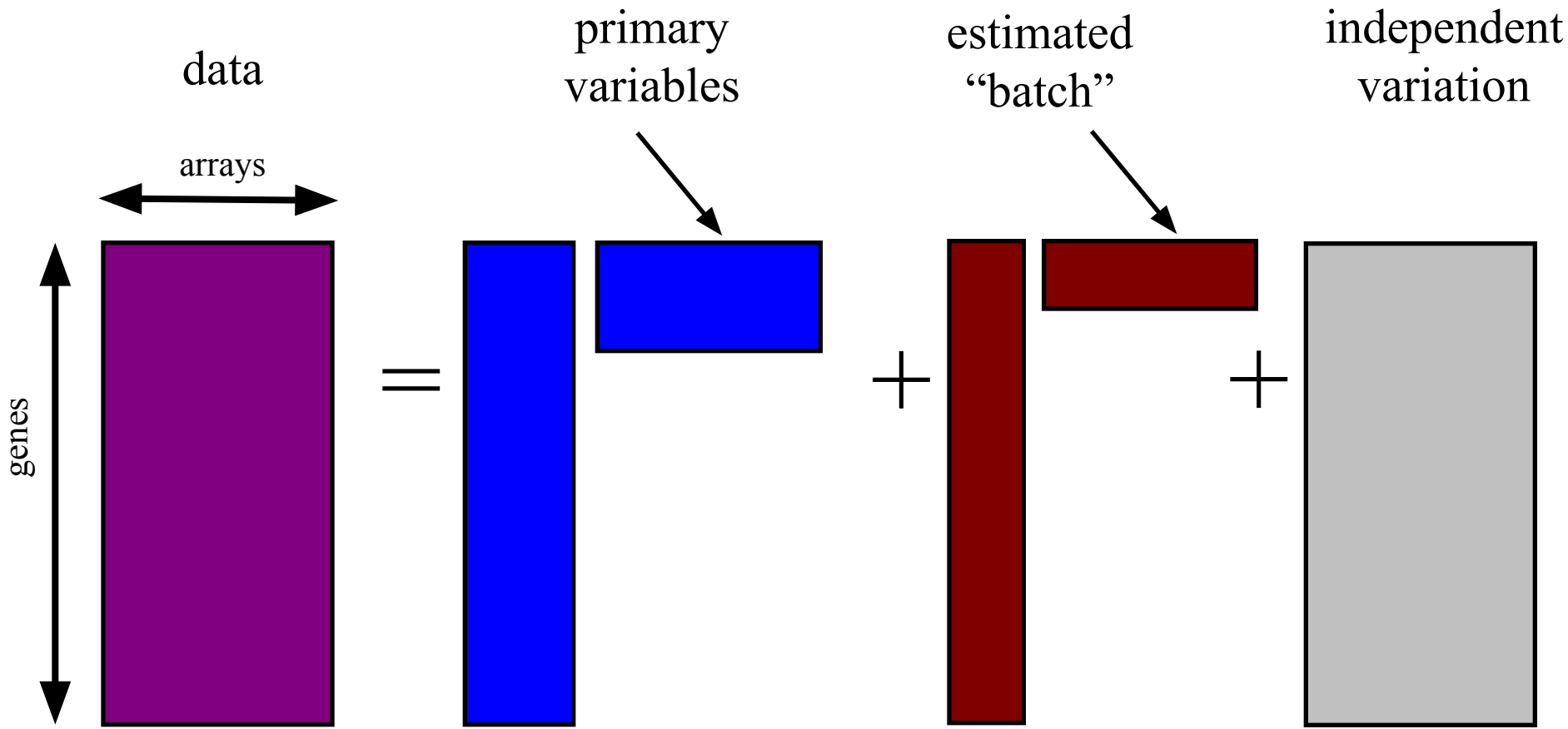


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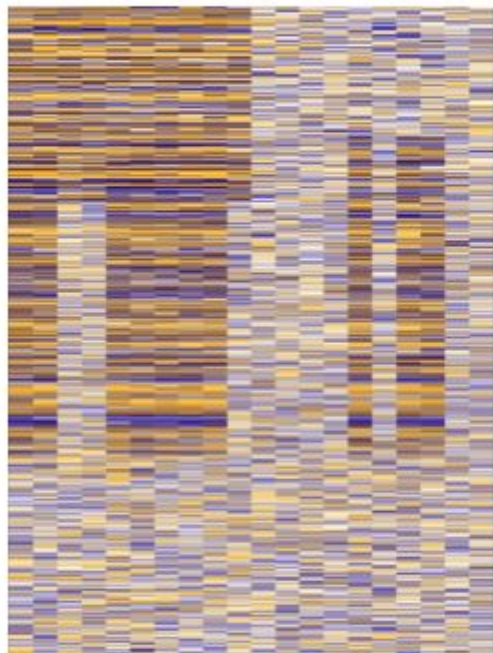
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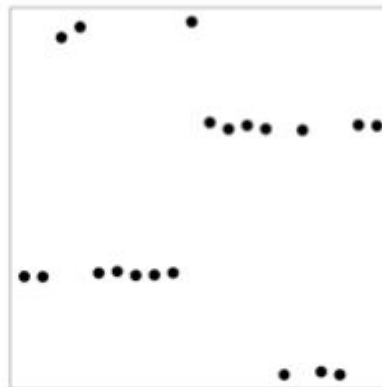
Surrogate Variable Analysis

The Data

$\Pr(\text{Group} \& \text{Batch})$



Estimate of Batch



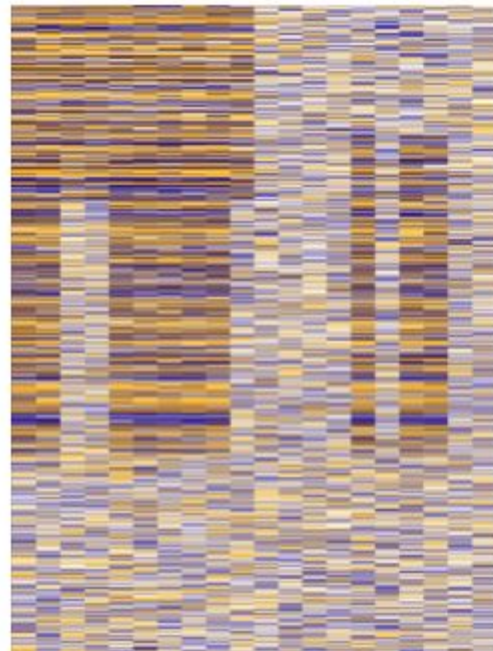
True Batch



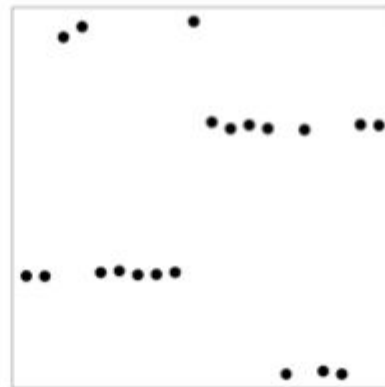
Surrogate Variable Analysis

The Data

$\Pr(\text{!Group} \ \& \ \text{Batch})$



Estimate of **Batch**



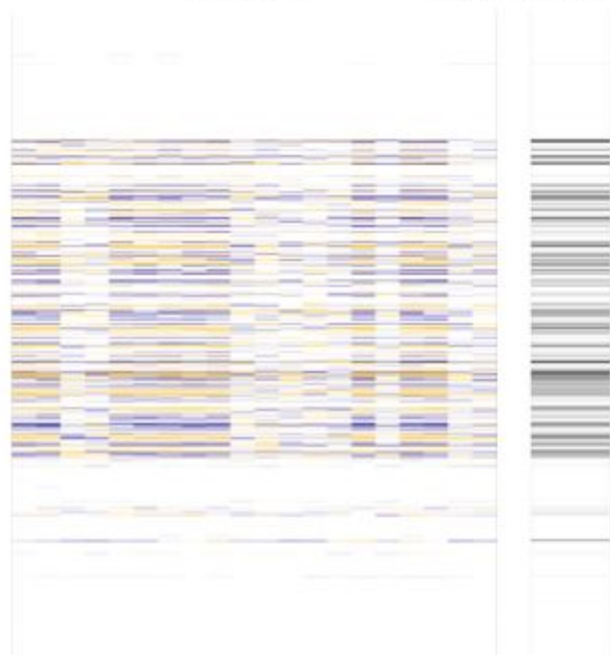
True **Batch**



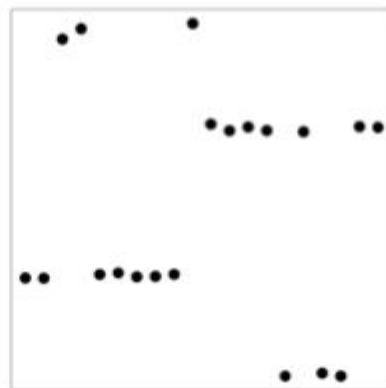
Surrogate Variable Analysis

The Data

$\Pr(\text{!Group \& Batch})$



Estimate of **Batch**



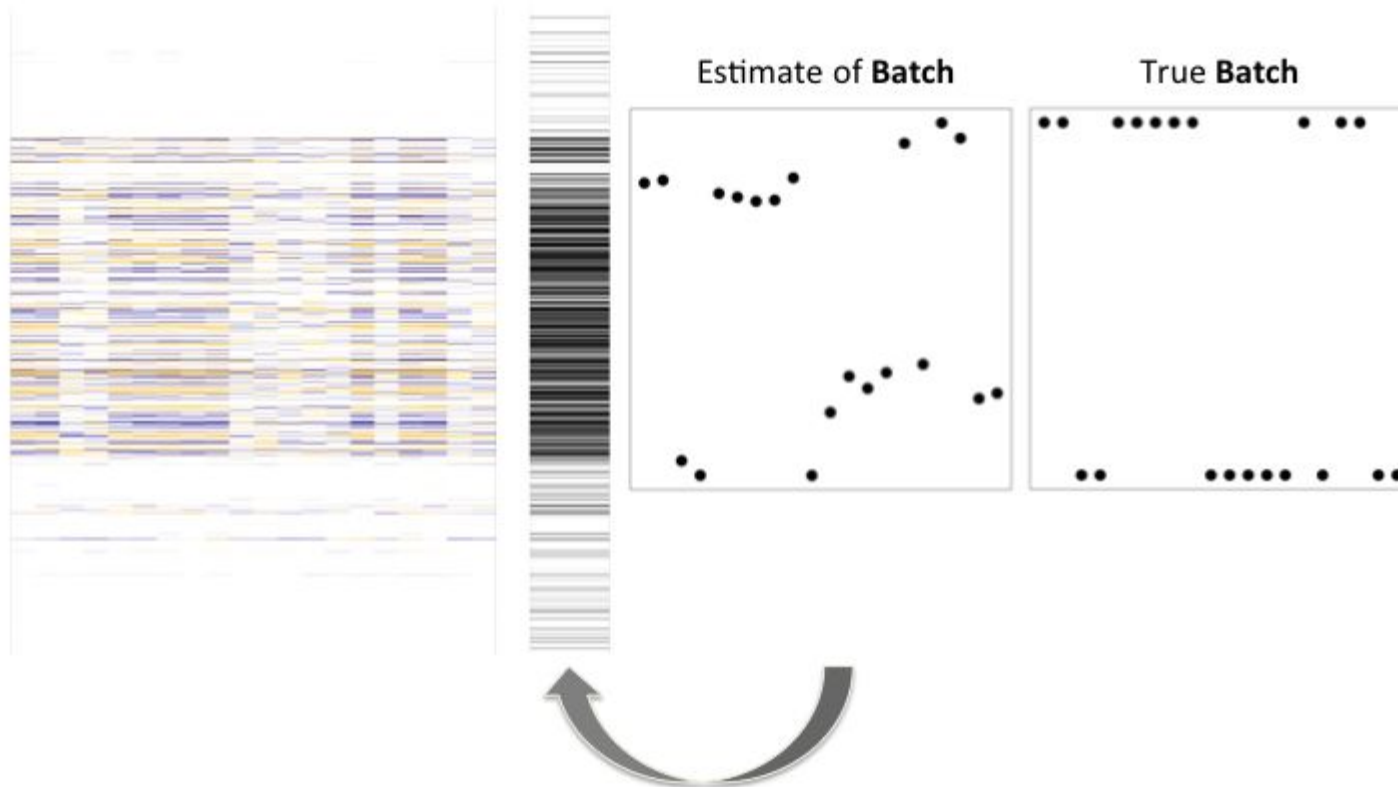
True **Batch**



Surrogate Variable Analysis

The Data

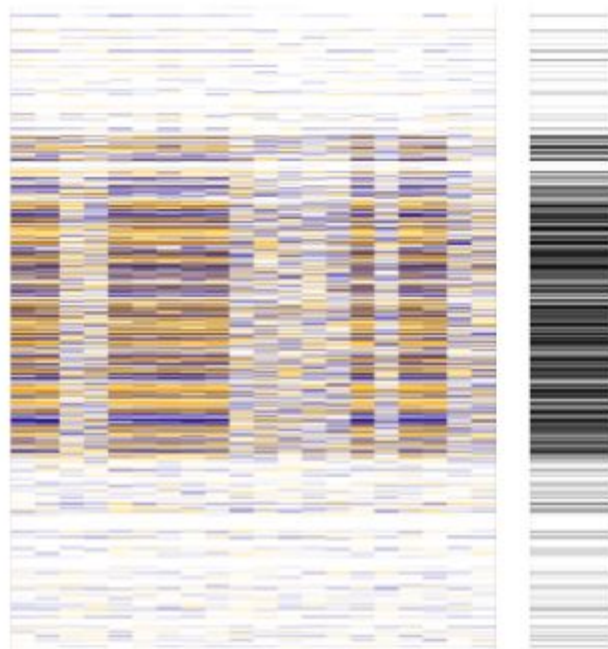
$\Pr(\text{Group \& Batch})$



Surrogate Variable Analysis

The Data

$\Pr(!\text{Group} \& \text{Batch})$



Estimate of **Batch**



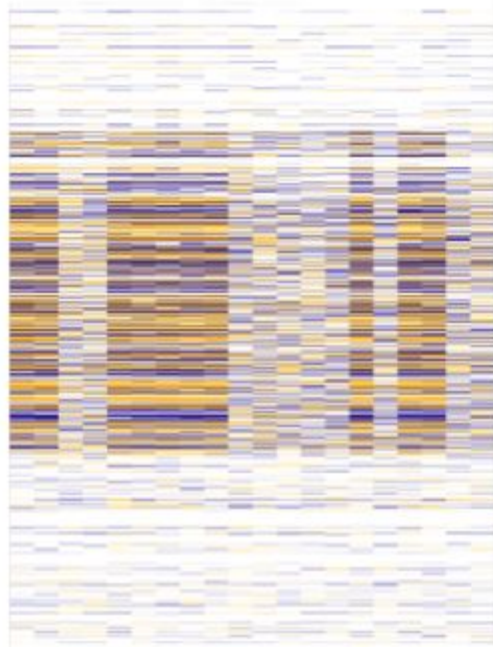
True **Batch**



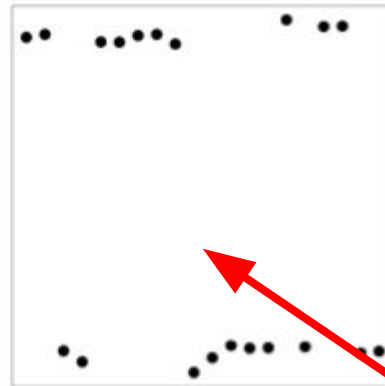
Surrogate Variable Analysis

The Data

$\Pr(\text{!Group \& Batch})$



Estimate of Batch



True Batch



Adjust for this!



Notes and further reading

- Introduction to batch effects
 - <http://www.nature.com/nrg/journal/v11/n10/abs/nrg2825.html>
- Introduction to batch effects with linear models
 - <http://biostatistics.oxfordjournals.org/content/8/1/118.abstract>
- Surrogate variable analysis
 - <http://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.0030161>